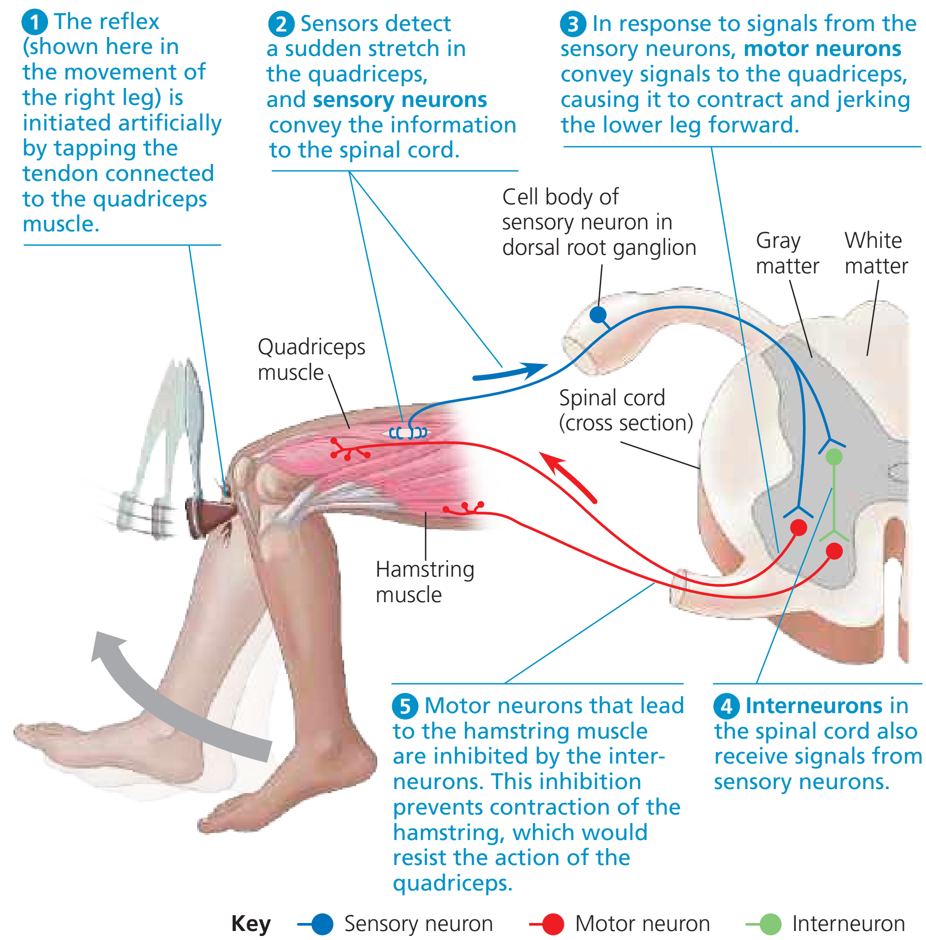
▶ **Figure 49.5 The knee-jerk reflex.** Many neurons are involved in this reflex, but for simplicity only a few neurons are shown.

MAKE CONNECTIONS *Using the nerve signals to the hamstring and quadriceps in this reflex as an example, propose a model for regulation of smooth muscle activity in the esophagus during the swallowing reflex (see Figure 41.9).*



During a physical exam, your doctor may trigger the knee-jerk reflex with a triangular mallet to help assess nervous system function **(Figure 49.5)**.

The Peripheral Nervous System

The PNS transmits information to and from the CNS and plays a large role in regulating an animal's movement and its internal environment **(Figure 49.6)**. Sensory information reaches the CNS along PNS neurons designated as *afferent* (from the Latin, meaning "to carry toward"). Following information processing within the CNS, instructions then travel to muscles, glands, and endocrine cells along PNS neurons designated as *efferent* (from the Latin, meaning "to carry away"). Note that most nerves are bundles of both afferent and efferent neurons.

The PNS has two efferent components: the motor system and the autonomic nervous system (see Figure 49.8). The neurons of the **motor system** carry signals to skeletal muscles. Motor control can be voluntary, as when you raise your hand to ask a question, or involuntary, as in the knee-jerk reflex controlled by the spinal cord. In contrast, regulation of smooth and cardiac muscles by the **autonomic nervous system** is generally involuntary. The sympathetic and parasympathetic divisions of the autonomic nervous system regulate organs of the cardiovascular, excretory, and endocrine systems. A distinct network of neurons now known as the **enteric nervous system** exerts direct and partially independent control over the digestive tract, pancreas, and gallbladder.

Homeostasis often relies on cooperation between the motor and autonomic nervous systems. In response to a

▼ **Figure 49.6 Functional hierarchy of the vertebrate peripheral nervous system.**